

Answer Key — Biology Grade 10

Sub-Strand 1.2: Chemicals of Life • Lesson 3: Carbohydrates: Starch and Sugars

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
B	B	A	D	B	D	A

USE AFTER: OBSERVE PHASE — Test 1: iodine test for starch

Q1. A group adds 2-3 drops of iodine solution to some maize flour mixed with water. Which result shows that starch is present?

- A. The mixture turns brick-red
- B. The mixture turns blue-black**
- C. The mixture turns yellow-orange and stays clear
- D. The mixture turns green

Correct answer: B. Iodine turns blue-black when starch is present. Brick-red and green are Benedict's test colours for reducing sugar, not iodine results.

USE AFTER: OBSERVE PHASE — Test 2: Benedict's test for reducing sugars

Q2. A learner mixes 2 cm³ of mashed ripe banana solution with 2 cm³ of Benedict's solution and heats it in a hot water bath for 4 minutes. It turns orange to brick-red. What does this show?

- A. The banana contains starch
- B. The banana contains a lot of reducing sugar**
- C. The banana contains protein
- D. The banana contains no carbohydrate

Correct answer: B. Benedict's solution changes from blue through green, yellow and orange to brick-red as the amount of reducing sugar increases, so orange to brick-red means a lot of reducing sugar is present.

USE AFTER: EXPLAIN PHASE — Explain Step 1: group discussion on ugali being blue-black with iodine but blue with Benedict's

Q3. Ugali turns blue-black with iodine solution but stays blue when heated with Benedict's solution. What does this tell us about the carbohydrate in ugali before digestion?

- A. It contains starch but no reducing sugar such as glucose**
- B. It contains no carbohydrate
- C. It contains both starch and a lot of glucose
- D. It contains glucose but no starch

Correct answer: A. Blue-black with iodine shows starch is present. Benedict's staying blue shows there is no reducing sugar (glucose) in the ugali before digestion.

USE AFTER: EXPLAIN PHASE — Explain Step 2: bead or paper-chain model of starch breaking into glucose

Q4. Ugali has no reducing sugar, yet Wanjiru's blood glucose went up after she ate it. Which statement best explains this?

- A. Starch is already glucose, so it passes straight into the blood
- B. Beans turn into starch in the stomach, which raises the glucose
- C. Starch changes into protein, which then enters the blood as glucose
- D. Starch is made of many glucose units joined together, so it must be broken down into glucose before entering the blood**

Correct answer: D. Starch is a large carbohydrate made of many glucose units. It must be broken down into single glucose units (monosaccharides) before glucose can enter the blood.

USE AFTER: EXPLAIN PHASE — Explain Step 3: using the numbers from the graph to write the CER statement

Q5. Wanjiru's blood glucose concentration was 4.5 mmol/L before lunch and reached a peak of 8.0 mmol/L after 60 minutes. By how much did it rise, in mmol/L?

- A. 3.0
- B. 3.5**
- C. 12.5
- D. 4.5

Correct answer: B. Rise = peak minus starting value = $8.0 - 4.5 = 3.5$ mmol/L.

USE AFTER: DQB PHASE — DQB: sorting sticky notes into answered, partly answered and still to investigate

Q6. After today's food tests, the class DQB has these four questions. Which one can the class now answer using the test results?

- A. Does the body use only carbohydrates for energy?
- B. Why did Wanjiru's blood glucose fall after 60 minutes?
- C. Which enzyme breaks starch into glucose?
- D. Do ugali and maize flour contain starch?**

Correct answer: D. The blue-black iodine result for ugali and maize flour answers this directly. The other questions need further investigation in later lessons.

USE AFTER: MODEL BUILDING PHASE — Revising the class flow-diagram model: adding evidence to the 'Food chemicals' box

Q7. Which statement about the chain from ugali to blood glucose is supported by today's evidence and should be added to the 'food chemicals' box of the class model?

- A. Ugali contains starch, which is made of glucose units, so it can be a source of the glucose in the blood**
- B. Nyama choma gives a blue-black result with iodine, so it is the main source of starch in Wanjiru's lunch
- C. Ugali gives a brick-red result with Benedict's solution, so it already contains glucose
- D. Only sweet foods contain carbohydrates, so ugali cannot be a source of glucose

Correct answer: A. Ugali is blue-black with iodine, so it contains starch, and starch is made of glucose units. Ugali stays blue with Benedict's, and nyama choma does not give a positive starch test.